📄 PHASE 5 – Project Flow & System Workflow

Project: Prism IQ

Version: 1.0

Status: Architecture Freeze

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1. System Overview

Prism IQ follows a **pipeline-driven architecture**, where every dataset passes through a fixed sequence of processing stages.

Instead of treating Upload, Cleaning, EDA, Machine Learning, and Reporting as separate features, they are connected into a single intelligent workflow.

```
Raw Data
     |
     ▼
Import
     |
     ▼
Validation
     |
     ▼
Profiling
     |
     ▼
Cleaning
     |
     ▼
EDA
     |
```

This ensures that every stage builds upon the validated output of the previous stage.

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2. Complete User Journey

```
Landing Page
       |
       ▼
Choose Import Method
```

```
      |
      ├── CSV
      ├── Excel
      ├── API
      └── URL
      |
      ▼
Dataset Validation
      |
      ▼
Dataset Successfully Imported
      |
      ▼
```

3. Data Import Flow

Prism IQ supports four import methods.

```
User
   |
   ▼
Choose Source
   |
   ├── CSV
   ├── Excel
   ├── API
   └── URL
         |
         ▼
 Import Service
         |
         ▼
Validation Engine
         |
         ▼
Create Dataset Metadata
```

Validation Rules

Every imported dataset is validated before entering the analytics pipeline.

Validation checks include:

- Supported file format

- File size
- Empty dataset
- Duplicate column names
- Invalid headers
- Unsupported data types
- API response validation
- HTML table extraction success

If validation fails, the user receives descriptive error messages without stopping the application.

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4. Data Profiling Flow

Immediately after successful import, profiling begins automatically.

```
Dataset
      |
      ▼
Profiling Engine
      |
      ├── Shape
      ├── Missing Values
      ├── Duplicate Rows
      ├── Data Types
      ├── Memory Usage
      ├── Unique Values
      ├── Statistics
      └── Quality Score
      |
      ▼
Store Metadata
      |
```

Output:

- Dataset Summary
- Quality Score
- Column Summary
- Initial Insights

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5. Smart Cleaning Flow

The cleaning engine performs rule-based preprocessing.

```
Profile
    |
    ▼
Cleaning Rules
    |
    ├── Missing Values
    ├── Duplicate Rows
    ├── Invalid Types
    ├── Category Standardization
    └── Optional Outlier Handling
            |
            ▼
Generate Preview
            |
            ▼
User Approval
            |
```

The original dataset always remains unchanged.

The cleaned version is stored separately.

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6. Exploratory Data Analysis Flow

```
Clean Dataset
       |
       ▼
EDA Engine
       |
       ├── Distribution Analysis
       ├── Correlation Analysis
       ├── Trend Analysis
       ├── Numerical Statistics
       └── Categorical Statistics
       |
       ▼
Visualization Engine
       |
```

```
        ▼
Interactive Dashboard
```

Generated visualizations include:

- Histogram
- Scatter Plot
- Bar Chart
- Line Chart
- Box Plot
- Heatmap
- Correlation Matrix

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7. Predictive Analytics Flow

```
Clean Dataset
      │
      ▼
User Selects Target
      │
      ▼
Problem Detection
      │
      ├── Regression
      └── Classification
      │
      ▼
Preprocessing
      │
      ▼
Train/Test Split
      │
```

The platform automatically compares supported models and presents the best-performing one along with evaluation metrics.

---

8. Recommendation Engine Flow

Unlike LLM-based systems, the MVP uses a deterministic rule engine.

```
EDA Results
      |
      ▼
ML Results
      |
      ▼
Rule Engine
      |
      ├── Trend Detection
      ├── Correlation Rules
      ├── Data Quality Rules
      ├── Model Performance Rules
      └── Business Rules
            |
            ▼
Recommendations
```

Example outputs:

- High-performing product categories
- Customer segments requiring attention
- Strong positive or negative correlations
- Suggested business actions

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9. Report Generation Flow

```
Dataset
     |
     ▼
Cleaning Summary
     |
     ▼
EDA Summary
     |
     ▼
ML Results
     |
     ▼
Recommendations
     |
     ▼
```

```
Report Builder
      |
```

Each report includes:

- Executive Summary
- Dataset Overview
- Cleaning Summary
- Key Visualizations
- Model Performance
- Recommendations
- Conclusion

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10. Frontend–Backend Communication Flow

```
React Frontend
        |
 REST API Request
        |
        ▼
FastAPI Router
        |
        ▼
Service Layer
        |
        ▼
Repository Layer
        |
        ▼
PostgreSQL
```

For analytical operations:

```
Service Layer
        |
        ▼
Analytics Engine
        |
        ▼
Return JSON
```

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11. File Management Flow

```
User Upload
      │
      ▼
uploads/raw
      │
      ▼
Cleaning
      │
      ▼
uploads/cleaned
      │
      ▼
EDA
      │
      ▼
reports/
      │
```

Only metadata is stored in PostgreSQL.

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12. Error Handling Flow

```
Request
    │
    ▼
Validation
    │
 ┌──┴─────────────┐
 │                │
Valid          Invalid
 │                │
 ▼                ▼
Continue      Return Error
 │
 ▼
Log Event
 │
```

```
  ▼
```

Every API returns a standardized response format with status, message, data, and errors where applicable.

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13. End-to-End System Workflow

```
User
  |
  ▼
Import Dataset
  |
  ▼
Validation
  |
  ▼
Profiling
  |
  ▼
Cleaning
  |
  ▼
EDA
  |
```

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14. Background Processing Strategy (Future-Ready)

Current MVP executes processing synchronously for simplicity.

The architecture is designed so that long-running tasks (large dataset profiling, model training, report generation) can later be moved to asynchronous workers such as Celery + Redis without major architectural changes.

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15. Data Lifecycle

```
RAW DATA
      |
      ▼
Validated
      |
      ▼
```

```
      Profiled
          │
          ▼
      Cleaned
          │
          ▼
      Analyzed
          │
          ▼
      Predicted
          │
```

Every stage updates the dataset status, making progress traceable and simplifying debugging.

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16. Design Principles

- **Single Source of Truth:** Dataset metadata lives in PostgreSQL.
- **Immutable Raw Data:** Original uploads are never modified.
- **Pipeline Consistency:** Each stage consumes the output of the previous stage.
- **Loose Coupling:** Import, Cleaning, EDA, ML, and Reporting are independent modules.
- **Extensibility:** New importers, models, or report templates can be added without changing existing flows.

✅ Phase 5 Deliverables

By the end of this phase, the complete operational flow of Prism IQ is frozen:

- End-to-end user journey
- Data processing workflow
- Frontend ↔ Backend communication
- Machine learning pipeline
- Report generation flow
- Error handling strategy
- File lifecycle
- Future-ready processing architecture